Open Source Coding - OPSC7311
Portfolio of Evidence
Part 1 - Research

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Research

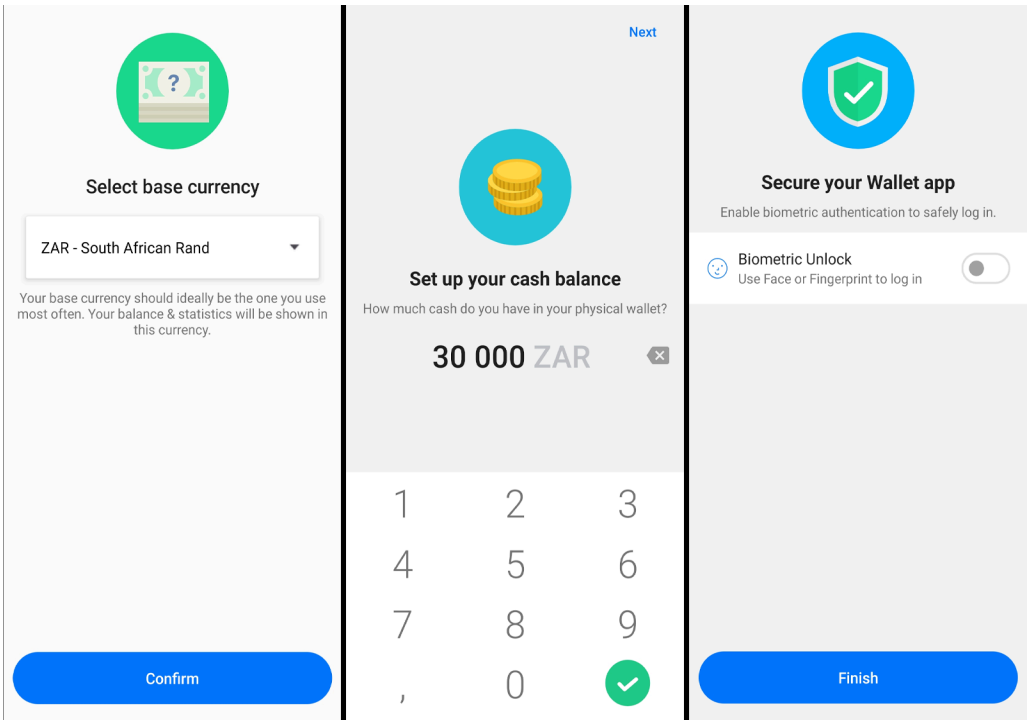
1. Introduction

As the world transitions to subscription models and the price to put food on the table and fuel in the car rises every month, South Africa's cost of living crisis can no longer be attributed to singular expenses but by the compounding effects of multiple different factors (Mawson, 2026). An efficient, effective and easy to use budgeting app is necessary to help keep track of all the different expenses a singular person or a family has. There will be an analysis of *Wallet* (Budget Bakers, 2026), *Vault22* (Vault 22, 2026) and *Goodbudget* (Dayspring Partners, 2026) to understand a trend of strengths and weaknesses of budgeting apps as well as to understand what users want out of a budgeting app.

2.1 Wallet

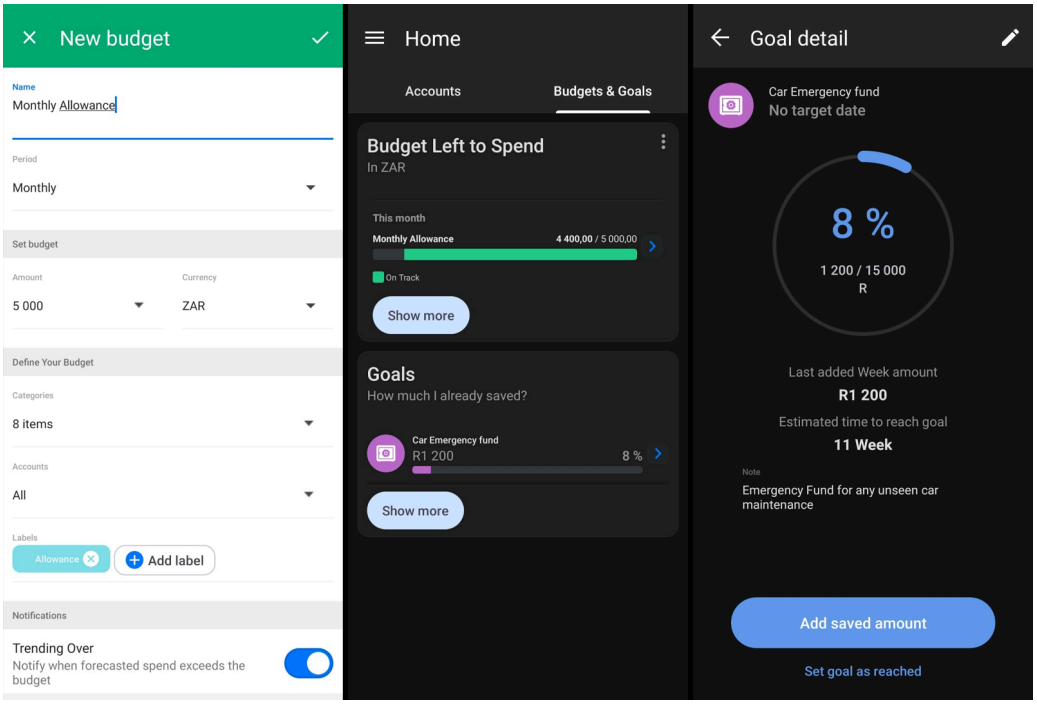
Overview

Wallet is a personal and family budgeting application that offers a suitable range of financial management features through a predominantly free-to-use model (Budget Bakers, 2026). Upon launching the application, users create an account, verify their email address, and configure initial settings such as their preferred currency and current bank balance. The application also offers biometric authentication, allowing only authorised users to access and modify their budget, improving security and protecting sensitive financial data (Budget Bakers, 2026).



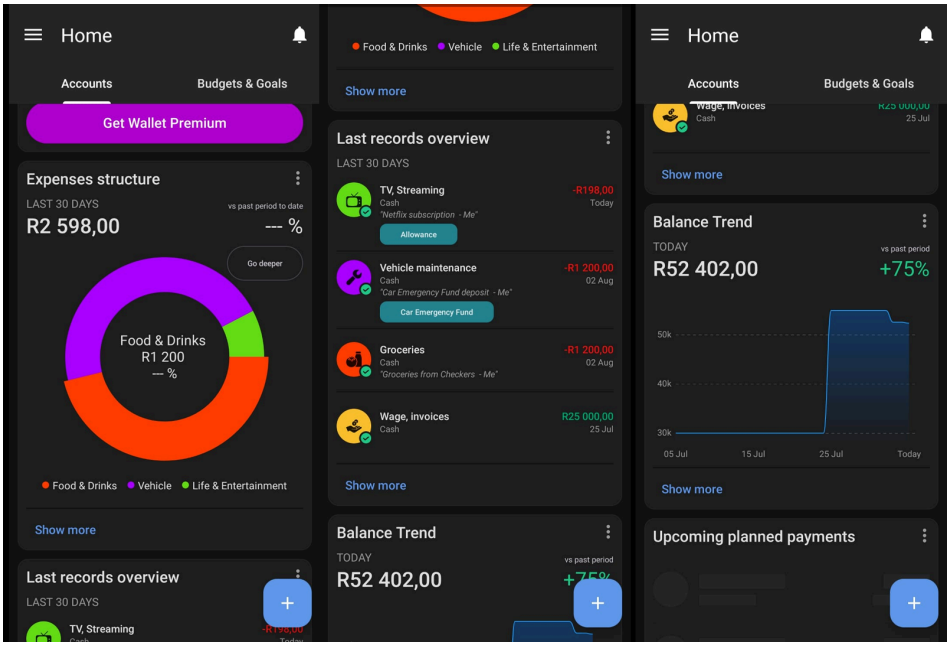
Screenshots from the initial process for beginning to track your budget (Budget Bakers, 2026).

Users can create weekly, monthly, or annual budgets with optional notifications as they approach spending limits. Budgets also calculate the maximum amount that can be spent each day while remaining within budget. Wallet further supports financial goals, enabling users to track progress towards objectives such as saving for a holiday, repaying loans, or covering future expenses.



Screenshots from the creating a new budget page, the Budgets & Goals page and the goal details page (Budget Bakers, 2026).

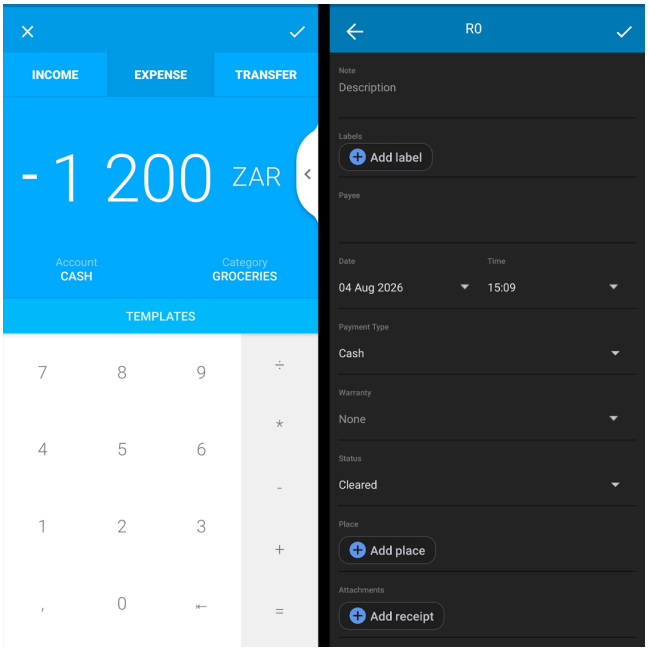
As financial data is recorded, Wallet generates visualisations including pie charts of monthly expenses and line graphs showing account balances over time, helping users interpret spending patterns and financial trends.



Screenshots from the home page showing expenses structure, a record overview and a balance trend (Budget Bakers, 2026).

Strengths and Weaknesses

Wallet has numerous strengths, reflected in its 4.6 out of 5 Google Play rating from approximately 375,000 reviews (Google Play, 2026). Recording income and expenses is straightforward, with optional receipt photos, additional transaction details, and reusable templates for recurring transactions (Budget Bakers, 2026).



Screenshots showing the pages for creating a new expense/income (Budget Bakers, 2026).

One of the application's primary weaknesses is its strict budgeting system. Expenses assigned to categories not included within a budget are excluded from its calculations, and there is no direct method to manually include or exclude specific transactions. This limits the flexibility and accuracy that some users may require.

Another limitation, common among budgeting applications, is reliance on user discipline. Research suggests that users often stop consistently using budgeting applications once the initial motivation declines (Sim, 2018). Unless automatic bank synchronisation is enabled, incomplete transaction records result in inaccurate analyses and budget tracking.

Although most user reviews are positive, some report dissatisfaction with the graphical user interface and persistent issues with the bank synchronisation feature (Google Play, 2026).

Innovative Features

A key innovative feature is Wallet's ability to synchronise with compatible bank accounts and payment services, automatically importing transactions and reducing reliance on manual data entry (Budget Bakers, 2026).

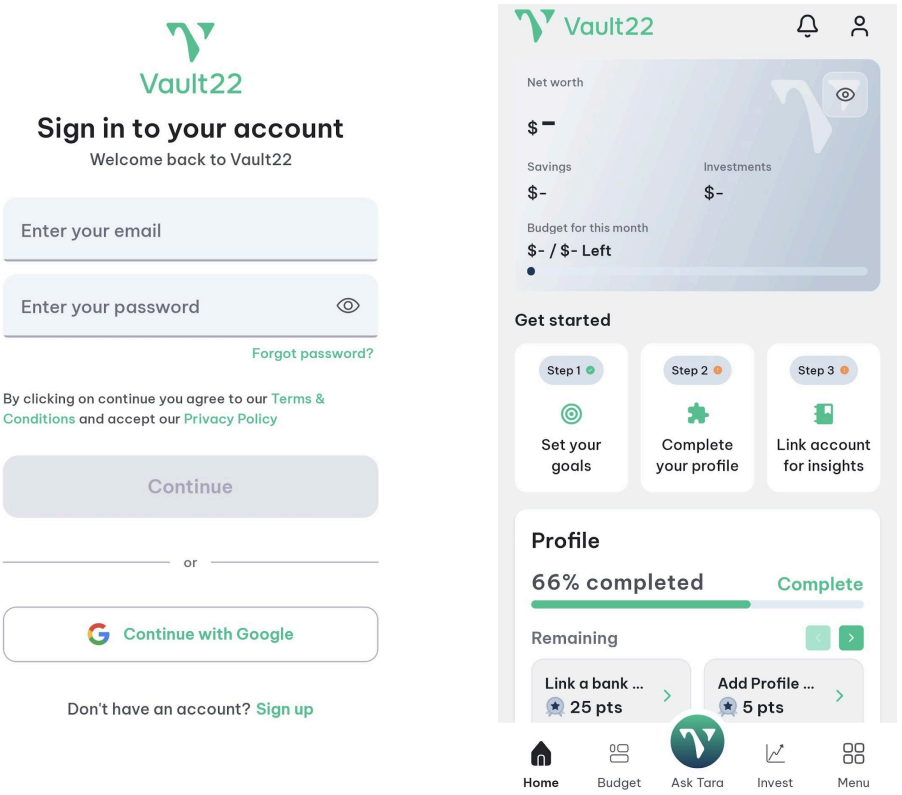
Wallet also supports biometric authentication using fingerprint or facial recognition, ensuring only authorised users can access or modify financial information. This provides an additional layer of security if a device is shared or lost.

2.2 Vault22

Overview

Vault22 is a financial wellness app, which was launched with backing from Standard Chartered's and Old Mutual's venture arms (Hypertext, 2024). The app gamifies finance management by allowing users to create goals that they can achieve and displaying them with progress bars as well as giving the player better titles for earning more savings or meeting their goals. The app has over 500 thousand downloads and a 2.4 star rating with over 4.5 thousand reviews.(Google Play, n.d.)

Vault22 also offers an investment feature, allowing users to invest directly into diversified ETF portfolios through the app as well as automated recurring investment contributions through debit orders. (Google Play, n.d.)



Picture of the sign in page and home page of Vault22(Vault22, 2026).

Strengths

- **Local market integration**

Unlike most budgeting apps, Vault22 is specifically connected to over 120 South African financial institutions. The app works directly in rand which means that users don't have to use workarounds to budget in their own currency unlike other apps that only work in USD for example. (Thorne, 2025)

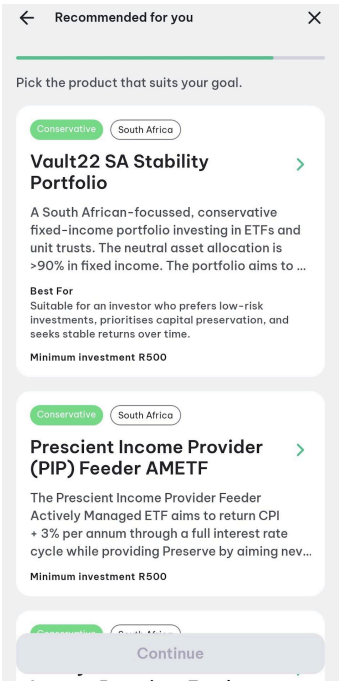
- **Free functionality**

The app's transaction history tracking and account aggregation as well as budgeting and goal tracking are free to use, as the app generates its income primarily from lending referrals and not user payments. (Google Play, n.d.)

Weaknesses

- **Customer service and reliability**

App store reviews suggest delays in customer service as well as frequent bugs (Google Play, 2026). For an app built around the idea of security and trust, this is a major issue as it breaks the promise of the app's branding. (RCS Group, 2025)



Picture of the in-app direct investments in Vault22 (Vault22, 2026).

Innovative features

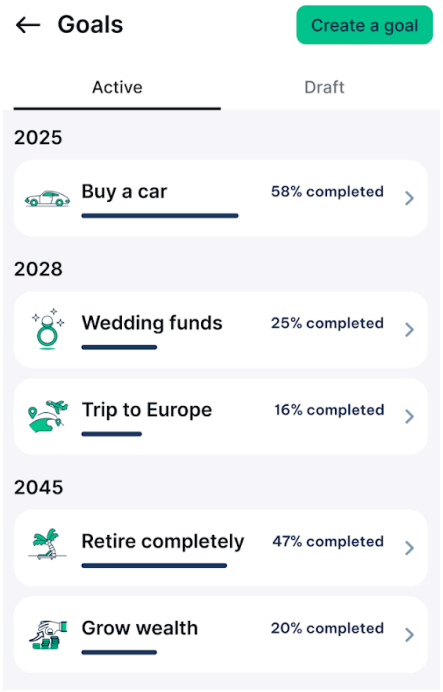
- **‘Nudges’**

The app gives personalised insights using social comparison, where the users spending is compared to similar peers, which functions as a behavioural nudge instead of just as a static report. This is more effective at changing user outcomes in spending, as it leverages social proof to pressure the user into improving their behaviour (Bignell, 2024).

- **Gamification**

The app gives users a ‘financial fitness level’ that can be increased in tiers from ‘Rookie’ to ‘Elite’ by improving your savings. The app also shows savings goals as progress bars for the user to track how close they are to meeting their goals. This borrows mechanics commonly

used in games to reward players for interacting with the systems by giving them rewards along the way and keeping them interested. (Schell, 2020)

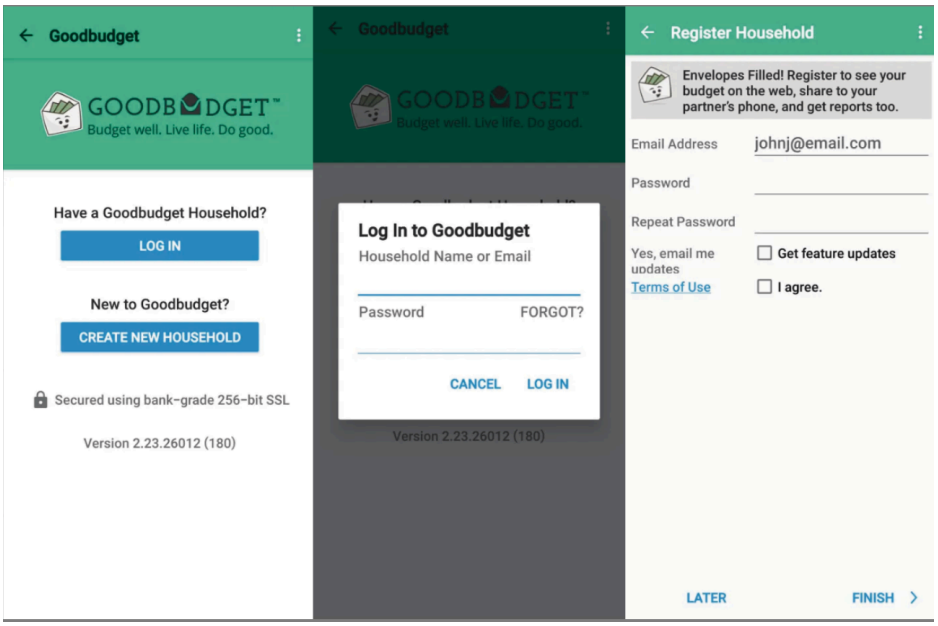


Picture of the goals progress bars in Vault22 (Vault22, 2026).

2.3 Goodbudget

Overview

Goodbudget is a budgeting app founded in 2009 in San Francisco. It modernizes the classic cash envelope method of money management, taking the typical budgeting experience online with virtual budgeting envelopes (Boyington, 2025). Instead of connecting a single account and automatically categorizing spending like most budgeting apps, Goodbudget lets users divide their income into “envelopes” categories like rent, groceries, and entertainment. Then tracks spending against those allocations, clearly showing funds available and warning when low or empty. It is also designed for couples or families to manage money jointly as it allows sharing budgets across multiple devices.



Screenshots from the login / account creation page. The login page and then the Register household page. (Goodbudget, 2026).

The app allows you to begin planning and budgeting very quickly in the free tier. Or to create an account that can be shared and connected to other budgets. Goodbudget is a planning-forward tool. It's more about planning finances than tracking past transactions (Boyington, 2025). It encourages allocating income rather than tracking expenses. Users can track debt payoff progress. Setting annual or long term savings goals through dedicated envelopes and view spending insights charts showing what percentage income goes towards each category.

Strengths

Many reviewers highlight Goodbudgets simplicity and its free tier. The free version offers up to 20 spending envelopes and access on two devices, making it usable by most people without paying for extra features (Barroso, 2026). Its simplicity also makes it approachable by those who aren't very tech savvy or experienced with budgeting apps. Its hands-on approach also helps users understand their spending habits. Goodbudget uses bank-rate encryptions and does not sell user data to third parties. Free tier users also never link their actual back accounts, so they never share login credentials with the app.

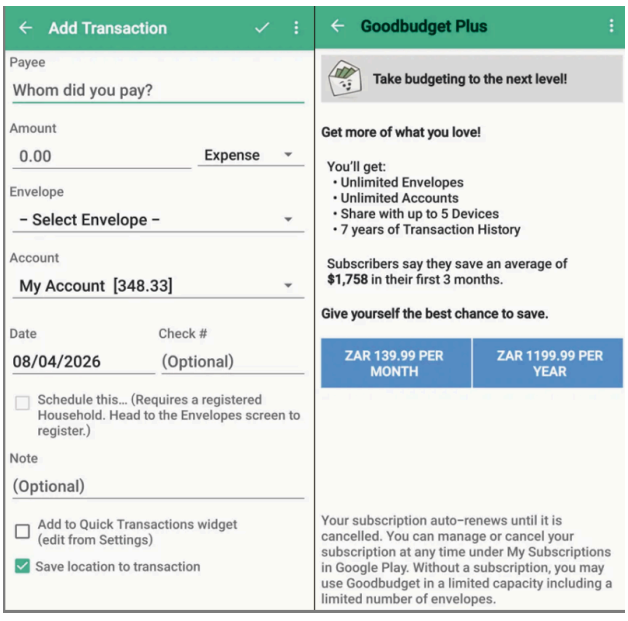


Screenshots from the envelopes page and the overall Spending details page (Goodbudget, 2026).

The concise interface makes use of graphics to visualize your current envelopes as well a pie chart to show overall spending. It also makes the information clear so you can read exactly what is going on within each envelope as well as the overall budget(Barroso, 2026).

Weaknesses

The manual nature that many find valuable is also its biggest drawback for people who want automatic budgeting (Barroso, 2026). The free version also doesn't connect to back accounts, meaning users must manually enter balances and transitions themselves. The app is also exclusively focused on budgeting and doesn't offer investment tracking, paying bills and other finance management tools. For full use users must either multiple apps or pay for Goodbudgets full package (Medine, 2026).



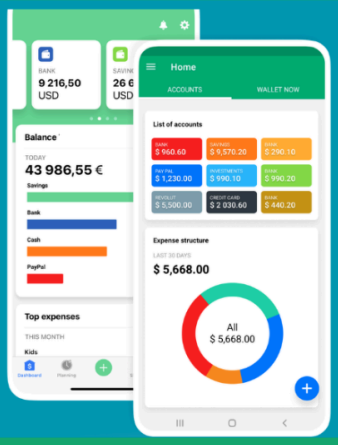
Screenshots from the add transaction page and the upgrade tiers page (Goodbudget, 2026).

Innovative features

Goodbudgets main innovation is not a single feature but more an idea. The platform emphasizes that budgets reveal priorities, the strength of shared planning and of giving (Medine, 2026). Newer additions to the paid tiers allow for manual and automatic merging, where multiple transactions can be matched into a single one, allowing for simpler and clearer record keeping (Boyington, 2025).

3. App Infographics

Wallet



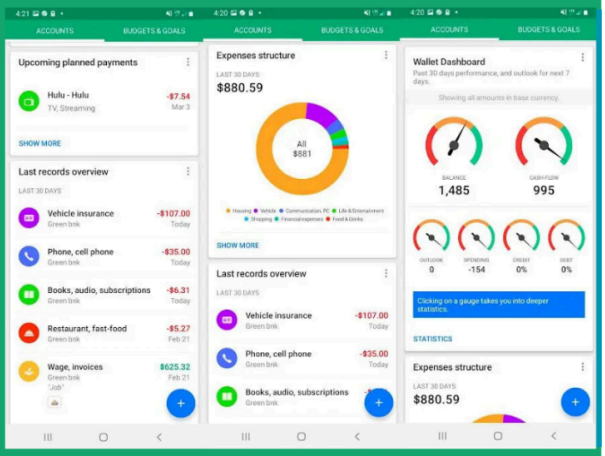
Overview

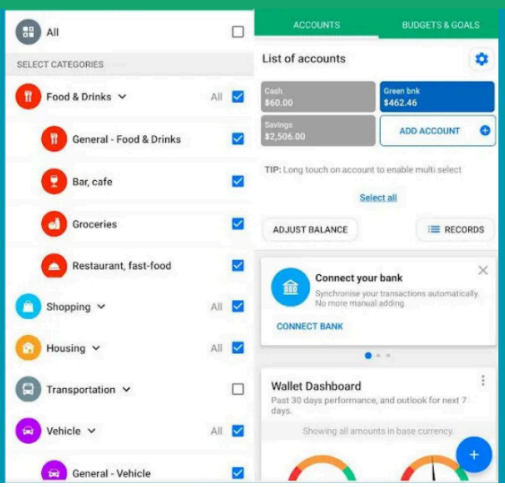
Wallet is a personal budgeting app that allows users to create a budgeting account to their preferred currency, and bank balance to create weekly, monthly or annual budget plans.
As data is recorded, Wallet helps interpret spending habits by providing graphs.

Strengths and Weaknesses

Strengths: Recording income and expenses is easy, with optional receipt photos, and templates for recurring transactions available.

Weaknesses: The budgeting system is strict. Expenses from non-included categories are excluded, and user discipline as the app is not synced with the bank transactions.

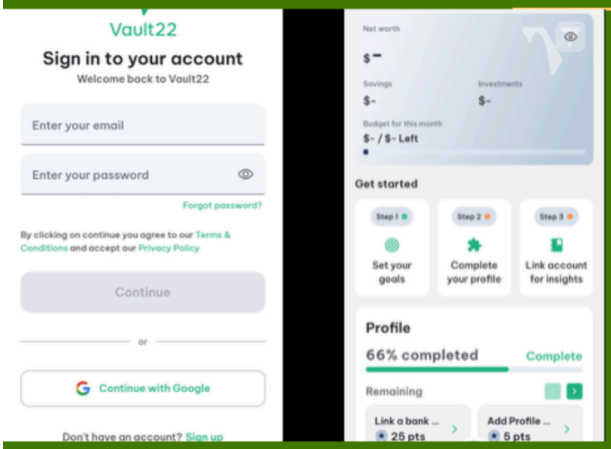




Innovative Features

Wallet is able to sync with bank accounts and payment services (only applies to compatible accounts). Wallet also supports biometric protection (fingerprint/face recognition) providing an extra layer of security.

VAULT22



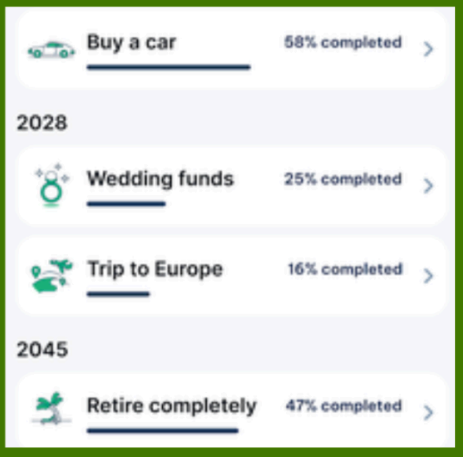
Overview

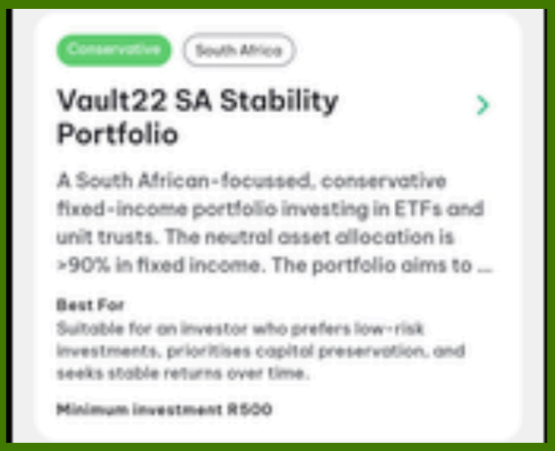
Vault22 is a budget app which gamifies financial management by allowing users to make goals which are displayed with progress bars, and giving better titles for earning more or meeting goals. Vault22 also has an investment feature, allowing direct investments into ETF portfolios.

Strengths & Weaknesses

Strengths: Vault22 is connected to over 120 financial institutions in South Africa, the app is free to use with no user payments.

Weaknesses: Delays in customer service and frequent bugs.



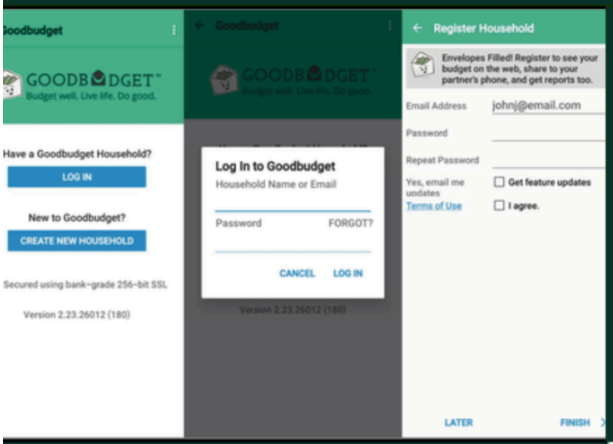


Innovative Features

Vault22 gives personalised insights through social comparison (comparing user spending to similar age peers) promoting better spending habits.

Vault22 gamifies finance management by saving goals as progress bars, serving as a reward system for players when complete.

Goodbudget



Overview

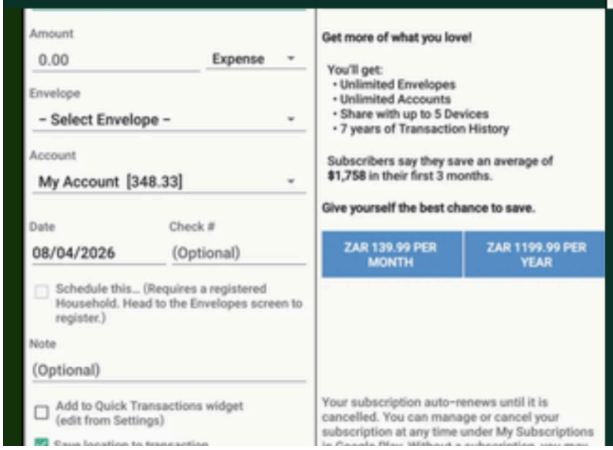
Goodbudget is a budgeting app that lets users divide their income into envelope categories and tracking the spending according to those categories, and warns when low and empty.
This encourages better allocations for income instead of tracking expenses.

Strengths & Weaknesses

Strengths: Free version offers up to 20 spending envelopes. Goodbudget is simple to use, and it uses bank-rate encryptions. Free users never link actual accounts.

Weakness: Manual nature for those who want auto budgeting, Free users must enter information themselves for a better experience. For full access users must pay.





Innovative Features

Goodbudget's main innovation is the idea that budgets reveal priorities, and the strength of shared planning and of giving.

New additions to the paid tiers allow for auto or manual merging and allow for a simpler and clearer record tracking.

4. Features we want to Include

- When a budget goal is set, there should be a consistent progress bar/report to track to give the user more incentive to budget better to complete their goal.
- Password/Access protection
- A system where a spending limit is set up (is primarily dependent on the budget period)
- Allow for short, medium, and long-term budget planning (weekly, monthly, yearly)

5. Conclusion

Ensuring financial wellness with budgeting apps is a sure-fire way to keep one's own funds in check, though a paywall is usually introduced when a user would be looking for more features to further aid their budgeting. Users of budgeting apps would like a free/predominantly free to use application like Wallet and Vault22 that can link to one's own bank account and track transactions to keep the user more consistently updated than always relying on the user to input data from transactions. Budgeting apps have already become a large part in helping people from all walks of life manage their finances better due to the cost of living crisis in South Africa. With the research conducted on the three budgeting applications, we have gone through strengths, weaknesses, and innovative features from each application which we will be keeping in mind when building our own budgeting app.

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